



Experiment 3

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Subject Name: Full Stack Development-II

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1. Aim:

To implement centralized state management in the EcoTrack application using Redux Toolkit and to handle asynchronous data operations using Redux async thunks with proper loading and error states.

2. Objective:

- To implement centralized state management in EcoTrack using Redux Toolkit.
- To manage asynchronous API calls using Redux async thunks.
- To handle loading, success, and error states effectively in the application.
- To structure Redux slices for scalable frontend architecture.
- To improve data flow and component communication through a global store.

3. Implementation/Code:

logsSlice.js:

```
import { createSlice, createAsyncThunk } from "@reduxjs/toolkit";
export const fetchLogs = createAsyncThunk(
  "logs/fetchLogs",
  async () => {
    await new Promise((resolve) => setTimeout(resolve, 1000));
    return [
      { id: 1, activity: "Car Travel", carbon: 4 },
      { id: 2, activity: "Electricity Usage", carbon: 6 },
      { id: 3, activity: "Cycling", carbon: 0 },
    ];
  }
);
```



```
);  
const logsSlice = createSlice({  
  name: "logs",  
  initialState: {  
    data: [],  
    status: "idle",  
    error: null,  
  },  
  reducers: {},  
  extraReducers: (builder) => {  
    builder  
      .addCase(fetchLogs.pending, (state) => {  
        state.status = "loading";  
      })  
      .addCase(fetchLogs.fulfilled, (state, action) => {  
        state.status = "succeeded";  
        state.data = action.payload;  
      })  
      .addCase(fetchLogs.rejected, (state, action) => {  
        state.status = "failed";  
        state.error = action.error.message;  
      });  
  },  
});  
export default logsSlice.reducer;
```

store.js:

```
import { configureStore } from "@reduxjs/toolkit";  
import logsReducer from "../logSlice.js";  
const store = configureStore({  
  reducer: {
```



```
logs: logsReducer,  
},  
});  
export default store;
```

Main.jsx:

```
import { StrictMode } from "react";  
import { createRoot } from "react-dom/client";  
import "./index.css";  
import App from "./App.jsx";  
import { Provider } from "react-redux";  
import store from "./store/store.js";  
import { AuthProvider } from "./context/AuthContext.jsx";  
createRoot(document.getElementById("root")).render(  
  <StrictMode>  
    <Provider store={store}>  
      <AuthProvider>  
        <App />  
      </AuthProvider>  
    </Provider>  
  </StrictMode>  
);
```

4. Output:

EcoTrack

Dashboard

Total Carbon Footprint: **12 kg CO₂**

- Car Travel - 1 kg CO₂
- Electricity Usage - 6 kg CO₂
- Cycling - 5 kg CO₂

High Carbon Activities (> 4)

- Electricity Usage = 6 kg CO₂
- Cycling = 5 kg CO₂

Low Carbon Activities (< 4)

- Car Travel = 1 kg CO₂



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5. Learning Outcome:

- Students will be able to configure Redux Toolkit store and slices.
- Students will understand how to use async thunks for API integration.
- Students will manage loading and error states in real-world React apps.
- Students will design clean and maintainable centralized state logic.
- Students will build production-ready data handling workflows in EcoTrack.